

MLink: 联合多个黑盒模型进行协作多模型推理

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模型推理的效率对于实际机器学习应用至关重要, 尤其是在延迟敏感的任务和资源有限的设备上。一个典型的困境是: 为了提供复杂的智能服务 (例如智慧城市), 我们需要多个机器学习模型的推理结果, 但是成本预算 (例如GPU内存) 不足以运行所有模型。在本工作中, 我们研究了黑盒机器学习模型之间的内在关系, 并提出了一种新的学习方法。任务: 模型链接 模型链接旨在通过学习不同黑盒模型之间的映射 (称为模型链接) 来弥合它们之间的知识。我们基于模型链接开发了一个调度算法, 名为MLink。通过模型链接实现多模态的协同推理, MLink可以在成本预算内提高推理结果的准确性。我们在具有7种不同ML模型的多模态数据集和具有6种ML模型的两个实际世界视频分析系统上评估了MLink。实验结果表明, 我们提出的方法可以在各种黑盒模型之间有效地构建模型链接。在预算范围内MLink可以节省667%的推理计算, 同时保持94%的推理精度, 优于多任务学习、基于深度强化学习的调度器和帧过滤基线

多模态推理任务越来越普遍, 例如智能音箱助手 (Bentley et al. 2018)、智慧城市 (Duan et al. 2018)、基于无人机的视频监控 (Dil-shad et al. 2020)、多模态自动驾驶 (Feng et al. 2020) 等。这些任务通常需要将来自多个模态的数据 (例如语音、图像、视频等) 结合起来进行分析和决策。为了应对这些工作负载, 研究人员已经提出了一系列多模态模型, 这些模型可以将来自不同模态的数据结合起来, 并从中提取有用的信息。这些模型通常使用深度学习技术来实现, 例如卷积神经网络 (CNN)、循环神经网络 (RNN) 和注意力机制 (Attention) 等。2020, etc Besides the accuracy of the trained model, the costs in the inference phase can become the bottleneck to the quality of services, especially for delay-sensitive tasks and resource-limited devices Towards cost-efficient inference, existing work explored various perspectives to achieve the resource-performance trade-offs Multi-task learning and zipping (He, Zhou, and Thiele 2018; Sanh, Wolf, and Ruder 2019; Crawshaw 2020; Zhang and Yang 2021) can reduce the computing overheads by sharing neurons among different tasks; Model compression (Hinton, Vinyals, and Dean 2015; Liu et al 2018; Gold-

Bai et al. (2020) and Blum et al. (2020) techniques aim to eliminate parameters and connections that are not relevant to the accuracy of inference. Inference reusing (Guo et al 2018; Ning, Guan, and Shen 2019) approaches aim to avoid repeating the same or similar computations; Source filtering (Li et al 2020) methods try to transmit only necessary input data to backend ML models Adaptive configuration (Jiang et al 2018) and multi-model scheduling (Yuan et al 2020a) were proposed to make inference workloads adaptive to the dynamics of input content. We summarize them as answers to an interesting question: How to obtain as accurate inference as possible with limited computing resources? In not executing machine learning models' outputs as the hint information We address this problem from a novel perspective: link-ing black-box models We were motivated by the insight that even ML models that are in input mode, learning tasks, architecture, etc. different types of machine learning models can share knowledge, because machine learning models are easy to show "overfitting" (Song and Shmatikov 2020), different models' outputs have semantic relevance (Tan et al 2018) If we can effectively share knowledge between different machine learning models, then we can directly predict the results of the remaining models. If this prediction cost is low, then we can reduce the cost budget, improve the accuracy of the results of the models that are not executed, and improve the accuracy of the results that cannot be obtained. To achieve this vision, we need to solve the following two main challenges: 1. How to effectively share knowledge between different machine learning models; 2. How to reduce the prediction cost. (1) How to build a link between machine learning models of different sizes and heights? In fact, the deployment of machine learning models can have different architectures and input modes, and it can be done by different programming languages and machine learning framework development. Heterogeneity

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